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QII Indicators

Results from Pilot Projects and Emerging Lessons

September 2023



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Executive Summary

The G20 Principles for Quality Infrastructure Investment (QII), endorsed by the G20 Finance Ministers and Central Bank Governors in June 2019, reflect the consensus that infrastructure is a driver of economic prosperity and provides a solid basis for strong, sustainable, balanced and inclusive growth and sustainable development. The Compendium of QII Indicators and the accompanying Guidance Note, endorsed by the G20 in June 2022, support the operationalization of the QII Principles by enabling the measurement of the contribution infrastructure projects make to economic, social, environmental, and development impact. To facilitate the implementation of the QII Indicators, the G20 asked IFC to pilot them on a set of projects voluntarily nominated for this purpose by G20 member countries. This Implementation Guide presents the results of this exercise.

The purpose of this document is to provide concrete guidance for countries on the end-to-end process of applying the QII Indicators to infrastructure projects and to offer insights on utilizing them to support strategic objectives around developing and mobilizing financing for quality infrastructure.

The following are the key lessons concerning the application of the QII Indicators:

1. QII Indicators proved to be broadly applicable across country and sector contexts and can be used to reflect a wide range of intended project outcomes.
2. Some QII indicators may be used to track alignment with more than one QII Principle and may be relevant to sectors other than those specified in the Compendium, depending on the project. Thus, while the Compendium presents indicators by QII Principle and sector, this categorization should be viewed as a guideline rather than a strict rule.
3. Data coverage is uneven across indicators with greater availability of data on operational and financial performance of projects aligned with Principles 1 and 2 and greater data coverage gaps in relation to other QII Principles, most notably Principle 4 (resilience).

Additionally, the pilot has generated the following lessons on the potential uses of QII indicators to improve project quality and mobilize private investment in infrastructure:

4. QII Indicators can help draw stakeholders' attention to performance in non-financial aspects of project quality, which tend to be subject to less measurement and monitoring, and provide a means to systematically track them.
5. Achieving alignment with the QII Principles need not entail the proliferation of project objectives – a narrowly focused project can be aligned with the QII Principles. In fact, an overly ambitious set of objectives can in some cases be counterproductive to achieving impact because it disperses the attention of stakeholders and makes monitoring more complex and costly.
6. QII Indicators can facilitate system-level planning and ensure alignment of infrastructure projects with the QII Principles over time when they form part of a common approach to incorporating quality considerations into project development.
7. The adoption of QII Indicators can pave the way for the diversification of instruments and sources of financing for infrastructure by improving data availability and setting realistic expectations regarding the results that can be achieved on a broad range of project outcomes, giving comfort to financiers regarding project quality, and creating feedback loops that help improve the quality of subsequent projects.

Introduction

This document presents preliminary results from a pilot exercise undertaken by the International Finance Corporation (IFC) to assess the application of Quality Infrastructure Investment (QII) Indicators to projects nominated by some G20 members. It also outlines some emerging lessons with respect to the application, as well as their potential role in maximizing the positive impacts of infrastructure investment. The Compendium of QII Indicators and the accompanying Guidance Note, endorsed by the G20 membership in July 2022, is a voluntary and non-binding tool for the operationalization of the QII Principles, consisting of a menu of commonly used indicators applicable to five infrastructure sectors: energy, water and wastewater, information and communications technology (ICT), and urban infrastructure. The indicators included in the Compendium have been gathered from existing indicators that are already in use by countries,¹ International Organizations, and Multilateral and Regional Development Banks, as well as from the Harmonized Indicators for Private Sector Operations (HIPSO), which is a set of harmonized indicators adopted by 28 development finance institutions. This pilot seeks to supplement the Compendium and Guidance Note by illustrating and clarifying the practical aspects regarding QII Indicators application to projects in different sectors, and at different points in their lifecycle. Projects were nominated by G20 members voluntarily, and the analysis conducted by IFC is based on documentation provided by G20 members as part of the nomination process. The list of projects analyzed to date, together with a sectoral breakdown, is illustrated in Figure 1 below.

Figure 1. Pilot projects sector coverage

Transportation	· Makassar-Parepare Railway · Nhan Tan Bridge / Noi Bai Airport Road · Samuel De Champlain Bridge · Mombasa Port · Port of Taranto
Water & wastewater	· São Simão* · Darwin Region Water Supply · Sarnese Vesuvian District Water Supply
Urban infrastructure - Urban mobility	· Delhi Mass Rapid Transit
Urban infrastructure - Solid waste	· São Simão*
Energy	· Salaria Castilian PV Plants
ICT	

* Project covers more than one sector

The overall objective of the pilot remains to select a set of indicators for each project that best reflects its objectives and alignment with QII Principles. The selection was conducted using a three-step process, as illustrated in Figure 2. The first step consisted of selecting a long list of potential indicators from the cross-cutting and sector-specific annexes of the Compendium, based on their relevance to the project objectives. The second step focused on minimizing costs associated with data collection and monitoring of indicators in one of two primary ways: by leveraging the project's existing reporting requirements and by limiting the total number of indicators to only the most relevant ones. The final step was to ensure that the resulting set of indicators captured the project's alignment

¹ This includes national and regional sources such as national development banks (e.g. BNDES and DBSA), national and subnational government agencies, ministries, regulators, and infrastructure operators.

with QII Principles in a balanced manner, drawing attention to project objectives and applicable QII Principles in proportion to their importance.

Figure 2. QII indicators selection process²



Once the QII Indicators for a project had been selected, the following attributes were defined for each indicator. A summary of applicable indicators for each project analyzed is provided in the next sections, whereas the tables presented in the Annex contain the detailed proposed sets of QII Indicators and their attributes for each pilot project.

Unit	A standard measurement unit for the indicator, considering the need for data comparability across projects
Source	Potential project-specific source of data for the indicator
Timing and frequency	When the indicator will be measured and how often
Historical values	Sample historical values available in project documents

In practice, the complete process of applying the indicators would require several further steps, as well as putting in place of systems and implementation capacity. The additional steps include developing customized and/or more detailed definitions of the indicators based on the generic definitions provided in the Compendium, including (if applicable) formulas to calculate indicator values and the sources of data for each input into the calculation. Baseline data would also be required for some quantitative indicators to measure changes over time, for example, in levels of access to infrastructure services or efficiency of operations. Finally, it would be necessary to define a process for collecting, verifying,³ and analyzing indicator data and potentially disclosing the results.

The requirements for systems and implementation capacity to support the process described above will depend on the intended end-use of the results by the country or other stakeholder wishing to implement the indicators and the volume of projects to which the QII Indicators would apply, but would, at a minimum, consist of the following. Firstly, the stakeholders' strategy for implementing QII indicators would need to be translated into a framework for indicator selection (e.g. similar to that presented in Figure 2). This is a one-time exercise with the possible need for refinements of the framework as the strategy evolves, or additional requirements are identified. Once the general framework has been established, it would be utilized to select indicators for each project, along with defining the relevant indicator attributes and the complete process for applying the indicators, as described above. This can be

² The indicator selection process and the resulting indicator sets for the pilot projects presented in this document represent one possible approach to applying the QII Indicators. In practice, countries are expected to use the QII Indicators on a voluntary basis and customize both the process of their application and the indicators themselves to their needs and circumstances.

³ The data collection process establishes how the reporting party will provide data in practice – for example, they may be required to fill out an online form or send digital copies of documentation. In some cases, a third party may be engaged to verify the accuracy of the information provided using an agreed-upon methodology.

done by professional(s) with knowledge of Monitoring & Evaluation methodologies, leveraging existing processes, data sources, and IT systems for collecting and managing project data and documentation (e.g. financial, operational, etc.). Additionally, the professional(s) in charge of implementing QII Indicators may benefit from having access to existing sector data collected by the government and collaboration with non-governmental institutions that produce relevant data and research. Resource planning for the implementation of QII indicators must consider the implementation strategy and existing capacity, as well as account for the need to ensure continuity to track progress on indicators over several years and eventually analyze the data to draw insights, communicate them to the relevant stakeholders, and iterate on the design of the QII Indicators implementation strategy and process.

São Simão Water, Sanitation, and Solid Waste Concession

Project overview

The concession project in the São Simão municipality in the state of Goiás in Brazil includes the modernization and expansion of water, wastewater, and solid waste systems, benefitting close to 20,000 people and bringing the municipality closer to achieving national targets for the universalization of these services by 2033. The investment plan includes construction of new water and wastewater pipes, installation of water meters, improvements to the municipality's existing landfill, construction of a new sanitary landfill, and the purchase of waste sorting equipment to safely handle municipal solid waste and increase recycling. The concession, structured by the municipality with the support from federal agencies, Caixa and SEPPI, was awarded to the Consórcio São Simão Saneamento through a competitive bidding process under Brazil's new legal framework for sanitation.

Project cost: R\$ 49 million (USD 9.80 million)

Sponsor: São Simão municipality

Private partner: Consórcio São Simão Saneamento

Status: Operational

QII Indicators selection

Identification of project objectives

The objective of investments in the water supply network is to ensure sufficient, reliable, and safe potable water provision to the population of the municipality for the next 35 years. The municipality plans to extend the water and wastewater network to 100% of households and to ensure adequate wastewater treatment capacity and quality to minimize health and environmental risks. One of the main challenges facing São Simão is the high per capita water usage, stemming from lack of metering and economic incentives to use water efficiently, and technical and commercial losses. The project will introduce water meters, as well as repair and extend the water supply network, aiming to increase coverage while also bringing per capita water usage down in line with similar municipalities. Investments in solid waste management aim to ensure that the municipality has adequate access to sanitary landfills for the duration of the concession and for several years after, as well as to minimize adverse impacts on the environment and human health. In line with these goals, the municipality is seeking to increase the share of municipal solid waste that is recycled, investing in better sorting facilities and equipment, as well as in providing decent working conditions for landfill workers. Finally, in the context of Brazil's aim to attract more private financing and expertise to reach its target of universal access to water and sanitation services, it is a priority for the government to ensure that concessions, such as the São Simão sanitation project, are well-structured and implemented in accordance with all applicable laws and regulations and international best practice. Table 1 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting, water and urban infrastructure annexes of the Compendium of QII Indicators.

Incorporating resource considerations

Data coverage for the project was assessed using public consultation documents and information on the concession available on the São Simão municipality's website.⁴ These contain historical data on access to, and quality of, water and sanitation services for the São Simão municipality, expected results of the project, and Key Performance Indicators (KPIs) for the concession. Indicators in these sources largely overlap with QII Indicators, as indicated in Table 1. Although QII Indicators related to resilience (Principle 4) were not found in project-specific documentation

⁴ São Simão Municipality, "Concorrência Pública 003/2021," <https://saosimao.go.gov.br/tp/concorrenca-publica-003-2021/>.

or third-party sources, their implementation would require only a slight change in methodology as the São Simão municipality already tracks highly related indicators.

Ensuring balanced representation of QII Principles

The São Simão project covers three distinct services, which need to be tracked using different sector-specific indicators for Principles 1, 2 and 3. Indicators related to Principles 4 and 6 are cross-cutting and are applicable to all three sectors, so they are fewer in number.

Table 1. QII indicator selection for the São Simão project

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Universalization of access to water, wastewater, and solid waste management services	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	- Population with access to safe drinking water* - Population with access to improved sanitation* - Proportion of municipal solid waste (MSW) collected and managed in controlled facilities* - Average period of water supply* - New connections to piped water and wastewater services* - Piped water pressure - People benefitting from improved solid waste management facilities
Job creation and improvement in incomes and working conditions (in solid waste management)		- Total employment effects (Tier 2) - Direct jobs supported by the project (Operations and Maintenance)* - Direct jobs created by the project (Operations and Maintenance)* - Technical and specialized skills for suppliers (Tier 2)
Reduction in per capita water use and losses	Principle 2: Raising economic efficiency in view of life-cycle cost	- Sustainable pricing mechanisms - Non-revenue water (commercial)* - Non-revenue water (technical)* - Additional capacity per currency invested*
Financial sustainability of sanitation services at affordable end-user tariffs		- Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)* - Value-for-money analysis conducted (Tier 2)*
Adequate wastewater treatment capacity and quality	Principle 3: Integrating environmental considerations in infrastructure investments	Wastewater treated*
Increase in share of waste recycled		Total weight of waste diverted from disposal and its recovery*
Reliability of water and wastewater services	Principle 4: Building resilience against natural disasters and other risks	- Preventive maintenance (Tier 2)⁵ - Ruptures in transportation and distribution pipes (Tier 2)⁶ - Fortuitous “incidents” affecting the wastewater collection network during dry weather (Tier 2) - Design improves resilience to failure of critical links (Tier 2)
Implementation of the concession in accordance with applicable laws and regulations	Principle 6: Strengthening infrastructure governance	- Information disclosure (Tier 2)* - Transparency and accountability measures in procurement and financial management supported in implementation (Tier 2)* - Fiscal sustainability (Tier 2)

* Indicator included in the public consultation documents

⁵ The Business Plan for the concession mentions that OPEX figures for the project include resources needed for maintenance. “Anexo X: Plano de Negócios Referencial,” 2021, p. 17-22.

⁶ The Concession KPIs include a related indicator, which measures the number of households affected by disruptions. “Anexo X: Indicadores de Qualidade e Desempenho,” 2021, p. 3.

Darwin Region Water Supply

Project overview

The Darwin Region Water Supply project is located in the Northern Territory of Australia and encompasses two components: (i) the return to service of Manton Dam and (ii) the construction of Adelaide River Off-Stream Water Storage (AROWS). The Manton Dam is an existing, out of service dam that can provide additional water supply in the short term (2025-27). The AROWS component is a new reservoir, which would provide a step change in water supply for the Darwin region in the medium term (2028 and beyond) with significantly lower environmental impacts than conventional instream dams. The project seeks to address the issue of insufficient water supply in the context of growing urban population and industrial activity, and opportunity to support high-value agricultural and horticultural production in the region. Darwin's water is currently sourced from the Darwin River dam and supplemented through groundwater resources, the use of which is unsustainable considering the region's demand growth and expected impacts of climate change.

Project cost: \$1,451.94 million

Sponsors: Northern Territory Government / Australian Government

Status: Pre-operational

QII Indicators selection

Identification of project objectives

The ultimate objective of the project is to provide safe, secure, reliable, and sustainable water supply to enable economic activity, support growth, and provide water security across the Darwin region over the long term. It seeks to do so by increasing the amount of water supplied to the Darwin region in a cost effective and efficient manner, while also minimizing adverse environmental and social impacts and addressing the following concerns regarding resilience of water supply: (i) the risk of ongoing reliance on a single source of water and lack of redundancy, (ii) vulnerability of water sources to climate change, and (iii) reliability, security and environmental risk associated with groundwater depletion. The Northern Territory government engaged a wide range of stakeholders to gather feedback and understand priority issues for each group during project development and committed to thorough assessment and monitoring of the implementation of this nationally significant, large-scale project. Table 2 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting and water and wastewater annexes of the Compendium of QII Indicators.

Incorporating resource considerations

Data coverage for the project was assessed based on the AROWS business case findings and the preliminary assessment report.⁷ Although the project is pre-operational, the baseline data and projections provided in these documents provide an idea of the data that can be expected to be collected once the project is operational. Further, some of the indicators related to the project design can already be assessed. As a large scale, complex, and high priority project for the government, the Darwin Region Water Supply project has been selected from among several alternatives based on thorough technical studies and analyses of its potential economic, environmental, and social impacts and risks.

⁷ Infrastructure Australia. "Darwin Region Water Supply Business case evaluation summary." PWC, "Adelaide River Off-stream Water Storage Project: Detailed Business Case Key Findings." Department of Trade, Business and Innovation, "Adelaide River Off-stream Water Storage - Part A: Preliminary Assessment," 2020.

Ensuring balanced representation of QII Principles

The Darwin Region Water Supply project consists of upstream investments that aim to alleviate future constraints on water resources with an emphasis on long-term economic, environmental, and social sustainability considerations aligned with Principles 2, 3, 4, and 5. The project's emphasis on governance is aligned with Principle 6. While the project is expected to ultimately contribute to growth and development of the Darwin Region, Principle 1 is not considered applicable because these impacts will be indirect, stemming from the elimination of a future constraint on water resources.

Table 2. QII indicator selection for the Darwin Region Water Supply project

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Select the most cost effective and efficient long-term water supply option based on comprehensive, whole-of-life economic analysis	Principle 2: Raising economic efficiency in view of life-cycle cost	· <i>Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)*</i> · <i>Value-for-money analysis conducted (Tier 2)*</i> · Additional capacity per currency invested · Sustainable pricing mechanisms* · Risk mitigation measurement (Tier 2) · Technological innovation (Tier 2)
Minimize adverse environmental and social impacts	Principle 3: Integrating environmental considerations in infrastructure investments	· <i>GHG Emissions reduced/avoided</i> · <i>Adopts an environmental and social mitigation hierarchy framework*</i> · <i>Provides tools, and data to improve E&S practices for stakeholder engagement (Tier 2)*</i> · Water consumption (conservation) (Tier 2) · Level of water stress (Tier 2)
Provide safe, reliable, and sustainable water supply, considering climate change impacts and water demand changes	Principle 4: Building resilience against natural disasters and other risks	· <i>Design durable to disaster risks (Tier 2)</i> · <i>Design improves resilience to failure of critical links (Tier 2)</i> · Budget committed to disaster and climate risk resilience measures (Tier 2) · Preventive maintenance (Tier 2)
Consider interests of diverse stakeholders in project design	Principle 5: Integrating social considerations in infrastructure investment	· <i>Stakeholder engagement*</i> · Design minimizes land acquisition and involuntary resettlement (Tier 2) · Infrastructure improvement in local community · Multi-generational equity (Tier 2)
Thoroughly assess and monitor implementation of a nationally significant, large-scale project	Principle 6: Strengthening infrastructure governance	· <i>Information disclosure (Tier 2)*</i> · <i>Transparency and accountability measures in procurement and financial management supported in implementation (Tier 2)</i> · Anti-corruption protocols & procedures (Tier 2) · Updated, timely shared, consistent, and comparable water information systems (Tier 2)

* Indicator included in the project's business case findings and the preliminary assessment report

Sarnese Vesuvian District Water Supply

Project overview

The project seeks to improve the water supply to 22 municipalities in Italy's Campania Region, which are experiencing high water losses of up to 60%, by facilitating in-depth analysis and monitoring of network performance, revising the existing district partitioning, improving control of the water pressure and technology to locate leaks, and replacing and rehabilitating sections of the network. It is managed by GORI, a mixed-capital company with the majority share owned by a consortium of the region's municipalities. The works will be executed in three phases. Phase 1 consists of mapping the water distribution network through a Geographic Information System (GIS) and examination of water resources, service interruptions, and water quality, leveraging an IT system developed as part of a 2015 program to digitalize the local water network. This will be followed by designing optimized water networks in Phase 2, and implementation of the optimized network, including pipe replacement, in Phase 3.

Project cost: EUR 50 million (USD 55 million)

Sponsor: GORI S.p.A.

Status: Pre-operational (in implementation with expected completion by the end of 2023)

QII Indicators selection

Identification of project objectives

The main objective of the project is to reduce water losses in the water distribution network by 42 percent from 74 million cubic meters as of 2020 to a target of 27 million cubic meters in 2024 and improve its performance, while ensuring that the project is executed efficiently and maximizes positive socio-economic impacts. It is expected to generate jobs in the local economy, reduce energy consumption and associated CO2 emissions, mitigate drought phenomena, and contribute to more sustainable use of water resources, among other benefits. These positive impacts are expected to be enabled by the continued digitalization of the water supply system.

Incorporating resource considerations

Information on availability of data for cross-cutting and water and wastewater sector QII Indicators, as well as expected indicator values for the project, was provided by the Italian Ministry of Economy and Finance and shows significant overlap with QII Indicators, as illustrated in Table 3.

Ensuring balanced representation of QII Principles

The objectives of the Sarnese Vesuvian District Water Supply project align with all six QII Principles. The project's primary objectives of reducing system losses and improving performance align with Principles 1 and 2, whereas the objectives with respect to how the project is designed and implemented map to the other Principles. Principle 1 reflects the expected boost to employment; Principle 3 reflects the project's commitment to reducing energy use and waste associated with construction; Principles 4 and 5 reflect the project's water resource sustainability and climate resilience benefits; and Principle 6 captures its contribution to advancing the digitalization of the water supply system, which helps achieve the design and operational efficiency underpinning all other outcomes.

Table 3. QII indicator selection for the Sarnese Vesuvian District Water Supply project

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Maximize socio-economic benefits in the implementation of the project	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	· <i>Total employment effects (Tier 2)*</i> · Direct jobs supported by the project (Operations and Maintenance) · Direct jobs created by the project (Operations and Maintenance)* · Construction jobs (temporary construction)*
Improve system performance		· <i>Piped water pressure (Tier 2)</i>
Reduce system losses	Principle 2: Raising economic efficiency in view of life-cycle cost	· <i>Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)*</i> · <i>Non-revenue water (technical)*</i> · Value-for-money analysis conducted (Tier 2)* · Non-revenue water (commercial)
Decrease energy consumption in the water network and associated CO2 emissions	Principle 3: Integrating environmental considerations in infrastructure investments	· <i>Energy consumption (conservation) (Tier 2)*</i> · <i>Level of water stress (Tier 2)</i> · GHG emissions reduction/avoided*
Reuse, recycle, or recover at least 70% of construction and demolition waste		· <i>Total weight of waste diverted from disposal and its recovery</i>
Contribute to sustainable use of water resources and resilience of supply	Principle 4: Building resilience against natural disasters and other risks	· <i>Ruptures in transportation and distribution pipes (Tier 2)*</i> · Preventive maintenance (Tier 2)* · Budget committed to disaster and climate risk resilience measures (Tier 2) · Design improves resilience to failure of critical links (Tier 2)
	Principle 5: Integrating Social Considerations in Infrastructure Investment	· <i>Multi-generational equity (Tier 2)*</i>
Continue to advance digitalization of the water supply system	Principle 6: Strengthening infrastructure governance	· <i>Updated, timely shared, consistent, and comparable water information systems (Tier 2)*</i>

* Historical or expected value for the indicator included in the project documents

Makassar-Parepare Railway

Project overview

The 146 km Makassar-Parepare railway line in South Sulawesi, Indonesia, is a greenfield project that will facilitate transportation of goods and passengers on the island of Sulawesi. It will contribute to growth of the national economy and the development of the island by supporting ecotourism, mining and industrial activity (e.g. Tonasa cement factory, special economic zone in Barru), as well as connectivity to Makassar, Garongkong, and Parepare ports and the Hasanuddin International Airport in Makassar. The first phase of the railway was developed using public funds, and the second was procured through a competitive bidding process under a PPP scheme, reaching financial close in 2021. The scope of the PPP includes the design, construction, financing, operation, and maintenance of a new segment of the railway and basic infrastructure, as well as the operation and maintenance of three existing segments of the railway.

Project cost: IDR 1,005 trillion (USD 68 million)

Sponsor: Ministry of Transportation of Indonesia

Private partner: PT. Celebes Railway Indonesia

Financing: Indonesian Government, PT Sarana Multi Infrastruktur (PT SMI), PT Indonesia Infrastructure Finance (IIF), Bank Syariah Indonesia (BSI)

Status: Operational since 2023

QII Indicators selection

Identification of project objectives

The primary objective of the rail project is to support Indonesia's sustainable development agenda by contributing to economic growth, creating job opportunities in South Sulawesi, and reducing greenhouse gas emissions by shifting freight and passenger transportation from road to rail. The second phase of the project was developed under a PPP scheme to enable private service provision and risk sharing. As Indonesia seeks to build momentum for greater private participation in infrastructure, the project represents an important opportunity to demonstrate the government's capacity to structure, tender, and successfully implement PPP projects. The project design sought to mitigate adverse impacts on the environment and local communities.

Table 4 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting and transportation annexes of the Compendium of QII Indicators.

Incorporating resource considerations

Assessment of the data coverage for the project relied on a presentation prepared for this pilot exercise by Indonesia, which included high-level information on project design and financing with limited detail on the data that informed project planning and results monitoring envisioned for the project's operational stage. The indicator selection considered indicators most commonly measured for railway projects, and which are unlikely to entail costly data collection efforts.

Ensuring balanced representation of QII Principles

The project objectives are aligned with five of the six QII principles with an emphasis on its contributions to the island's sustainable growth and development (Principle 1) and avoiding negative impacts on the environment (Principle 3) and stakeholders (Principle 5). The indicator selection also reflects the goals of attracting private participation, risk sharing, and good governance under the PPP model (Principles 1, 2, and 6, respectively).

Table 4. QII indicator selection for the Makassar-Parepare railway project

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Contribute to economic growth and job creation in South Sulawesi	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	• <i>Total employment effects (Tier 2)</i> • Time between contract signature and financial close
Facilitate transport of goods and passengers		• <i>Passenger and freight volumes by mode of transport*</i> • Direct jobs created by the project (Operations and Maintenance) • Construction jobs (Temporary Construction) • Journey times (Tier 2)
Maximize efficiency of use of public funds and transfer portion of the risk to the private sector		• <i>Private capital mobilization*</i> • <i>Additional capacity per currency invested</i> • <i>Risk mitigation measurement (Tier 2)</i> • <i>Value-for-money analysis conducted (Tier 2)*</i> • Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2) • Operation & Maintenance cost recovery from fees / user tariffs (Tier 2)
Provide a more sustainable alternative to road transportation	Principle 3: Integrating environmental considerations in infrastructure investments	• <i>GHG emissions reduced / avoided</i> • Local air pollutants reduced (Tier 2) • Energy consumption (conservation) (Tier 2) • Adopts an environmental and social mitigation hierarchy framework • Contribution to country's transport sector NDC target (Tier 2)
Maximize benefits and minimize adverse impacts on affected communities	Principle 5: Integrating social considerations in infrastructure investment	• <i>Stakeholder engagement</i> • <i>Design minimizes land acquisition and involuntary resettlement (Tier 2)</i> • Compensation to displaced inhabitants (Tier 2) • Supports fair and full access to transportation services and economic opportunities by marginalized regions (e.g., rural) (Tier 2)
Ensure transparency and proper use of funds	Principle 6: Strengthening infrastructure governance	• <i>Transparency and accountability measures in procurement and financial management supported in implementation (Tier 2)</i> • <i>Anti-corruption protocols & procedures (Tier 2)*</i> • <i>Information disclosure (Tier 2)*</i> • Fiscal sustainability (Tier 2) • Introduction of new type of concession terms (Tier 2) • Information transparency & access (Tier 2)

* Indicator cited in the project documents

Delhi Mass Rapid Transport System Phase 2

Project overview

The Delhi Mass Rapid Transport System (Delhi MRTS) Project Phase 2, completed in 2011, was designed to cope with the growing road traffic in the Delhi metropolitan area, the capital city of India, by extending six metro lines and adding a total of 83 km and 734 coaches to the metro network. At the time of appraisal, India was rapidly urbanizing, and the share of the population owning vehicles and two-wheelers was growing at 12% per annum, whereas public transport use was declining. The project aimed to contribute to regional economic development and improve the urban environment by alleviating traffic congestion and reducing air pollution from vehicles.

Project cost: JPY 379,866 million (USD 3,139 million)

Sponsor: Delhi Metro Rail Corporation

Financing: 55% of the project cost was financed using a Japanese ODA loan to the Delhi Metro Rail Corporation Ltd. (DMRC), with the rest provided by the Government of India, the governments of National Capital Territory of Delhi and Haryana State and DMRC.

Status: Operational, completed in August 2011

QII Indicators selection

Identification of project objectives

The expected impacts of the Delhi MRTS project, as stated in the evaluation by the Japanese ODA, were “the regional economic development and improvement of the urban environment through alleviation of traffic congestion and reduction of traffic pollution,” which the evaluator further broke down into their components as follows.

Improvement of urban environment

- a. Reducing traffic congestion
- b. Saving travel time
- c. Reducing road accidents
- d. Reducing energy consumption
- e. Reducing air pollution
- f. Reducing noise pollution

Regional economic development

- g. Increase in property value along the metro corridor
- h. Increased business activities along the metro corridor

The project aimed to support regional economic development and reduce private vehicle use by providing Delhi residents with a convenient, reliable, and safe public transportation, which would increase accessibility of certain parts of the city, boosting property values and business activity. The project took measures to ensure that the metro would receive adequate maintenance and be operated by well-qualified employees, who receive training at a specialized institute that prepares professionals not only to be employed in the Delhi metro but also in other cities in India and the South Asia region. In order to address some of the shortfalls of the older metro lines, the project also incorporated social inclusion considerations, particularly women’s safety and access for people with disabilities. Table 5 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting and urban infrastructure annexes of the Compendium of QII Indicators.

Incorporating resource considerations

The Delhi MRTS project was subject to Japanese ODA reporting requirements as well as an ex-post evaluation four years after the project’s commercial operation date. The evaluation included an assessment of a wide range of

outcomes with significant overlap with QII Indicators, as illustrated in Table 5. The table distinguishes between indicators for which measurements were taken specifically for the project from those that rely on estimates and extrapolation from city-level data made during the ex-post evaluation. In case of the latter, application of these indicators to the project would potentially require a change in methodology to enable more precise measurement.

Ensuring balanced representation of QII Principles

Table 2 illustrates that in the case of Delhi MRTS, the number of applicable indicators and data availability are greater for Principles 1 and 2. A balanced set of indicators will, therefore, be able to leverage existing reporting, but require stakeholder to establish processes to collect project-specific indicator data to track Principles 3, 4, and 5.

Table 5. QII indicator selection for the Delhi MRTS project

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Reduce traffic congestion and travel time	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	· <i>Average daily ridership*</i> · <i>Decreased journey times**</i> · <i>Journey experience (Tier 2)*</i> · Vehicle average speed (urban) (Tier 2)
Ensure efficient operation of the transport system	Principle 2: Raising economic efficiency in view of life-cycle cost	· <i>Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)*</i> · Load factor · Additional capacity per currency invested* · Value-for-money analysis conducted (Tier 2)
Reduce energy consumption, air pollution, and noise pollution	Principle 3: Integrating environmental considerations in infrastructure investments	· <i>GHG Emissions reduced/avoided**</i> · <i>Local air pollutants reduced (Tier 2)**</i> · <i>Measures to limit noise pollution (Tier 2)*</i> · Energy consumption (conservation) (Tier 2)**
Provide reliable service	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	· <i>Development of employee skills (Tier 2)</i> · Service frequency*
	Principle 4: Building resilience against natural disasters and other risks	· <i>Diligent maintenance and continuity (Tier 2)*</i>
Ensure that public transit is convenient for all passengers	Principle 5: Integrating social considerations in infrastructure investment	· <i>Usability of public transport, by gender (Tier 2)</i> · <i>Proportion of population that has convenient access to public transport, by sex, age, and persons with disabilities (Tier 2)</i> · Design minimizes land acquisition and involuntary resettlement (Tier 2) · Disability & special needs integration (Tier 2)

* Project-specific historical data available

** Estimated as part of an ex-post evaluation

Nhat Tan Bridge and Noi Bai International Airport to Nhat Tan Bridge Connecting Road

Project overview

The project encompasses the Nhat Tan Bridge (Vietnam-Japan Friendship Bridge), which passes over the Red River flowing through Hanoi City and approach roads, and the Noi Bai International Airport to Nhat Tan Bridge Connecting Road, which enhances transportation capacity from central Hanoi to the airport. The project was implemented in response to severe traffic congestion that emerged in the early 2000s as a result of rapid growth and insufficient road network and inadequate road width. With few bridges over the Red River, cargo vehicles often had no convenient option to go around Hanoi and instead had to pass through the city center. The existing North Thang Long - Noi Bai road was saturated, making access between the city center in the south and the Noi Bai International Airport, industrial parks, and newly developed areas in the north inefficient, further adding to the traffic problem. The Nhat Tan Bridge and Connecting Road helped address these deficiencies by allowing traffic to bypass the city center via the new bridge and providing a shorter and more direct route between the city and developments to the north via the new road.

Project cost: Nhat Tan Bridge: JPY 62,640 million (USD 575 million); Noi Bai International Airport to Nhat Tan Bridge Connecting Road: JPY 24,168 million (USD 222 million)

Sponsor: Government of Vietnam / Ministry of Transport

Financing: ODA Loans from Japan to the Government of Vietnam / Ministry of Transport financed 80% of the cost of the bridge and 62% of the connecting road.

Status: Operational since January 2015

QII Indicators selection

Identification of project objectives

The primary development objectives of the Nhat Tan Bridge and Connecting Road were to address the increasing traffic demand, streamline logistics, strengthen transportation capacity, and alleviate traffic congestion in the Hanoi area, ultimately contributing to economic growth and enhancing Vietnam's international competitiveness. Objectives with respect to the quality of project implementation included ensuring adequate supervision and maintenance standards, for which the Hanoi Department of Transport provided relevant training to Hanoi City prior to turning over the responsibility for operation and maintenance of the project. Other priorities included managing risks related to the natural environment, resettlements, and land acquisition in line with the relevant laws and regulations of Vietnam. Table 6 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting, transportation, and urban infrastructure annexes of the Compendium of QII Indicators.

Incorporating resource considerations

The project was subject to Japanese ODA reporting requirements as well as an ex-post evaluation four years after the commercial operation date. The evaluation, which was used to assess data coverage, included assessment of a wide range of outcomes with significant overlap with QII Indicators, as illustrated in Table 6.

Ensuring balanced representation of QII Principles

The project's objectives emphasize alignment with five of the six QII Principles. The goal of reducing congestion and associated benefits to stakeholders and the environment are captured under Principles 1, 2, and 3. Indicators under Principles 4 and 5, along with the indicator on employee skills development under Principle 1 capture key considerations for ensuring the long-term benefits of the project through risk mitigation and proper maintenance.

Table 6. QII indicator selection for the Nhat Tan Bridge and Connecting Road project

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Alleviate congestion and improve efficiency of logistics	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	· <i>Average daily traffic (or infrastructure utilization rate)*</i> · <i>Journey times*</i>
Ensure long-term benefits of the project through proper operation and maintenance	Principle 2: Raising economic efficiency in view of life-cycle cost	· <i>Development of employee skills (Tier 2)*</i> · <i>Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)*</i> · <i>Vehicle operating costs (roads/ports)*</i> · <i>Additional capacity per currency invested*</i> · <i>Value-for-money analysis conducted (Tier 2)*</i>
	Principle 4: Building resilience against natural disasters and other risks	· <i>Diligent maintenance and continuity (Tier 2)*</i>
Mitigate environmental and social risks associated with bridge and road construction	Principle 3: Integrating environmental considerations in infrastructure investments	· <i>GHG Emissions reduced/avoided</i> · <i>Adopts an environmental and social mitigation hierarchy framework*</i> · <i>Local air pollutants reduced (Tier 2)</i> · <i>Provides tools, and data to improve E&S practices for stakeholder engagement (Tier 2)</i> · <i>Measures to limit noise pollution (Tier 2)</i>
Provide an alternate route to the airport in the event of a disaster	Principle 4: Building resilience against natural disasters and other risks	· <i>Project introduces design or technical characteristics that enable operability in all weather conditions</i> · <i>Improvement in employee technical and disaster management skills (Tier 2)</i> · <i>Design durable to disaster risks (Tier 2)</i> · <i>Budget committed to disaster and climate risk resilience measures (Tier 2)</i>
Comply with relevant laws and regulations on resettlements and land acquisition	Principle 5: Integrating social considerations in infrastructure investment	· <i>Compensation to displaced inhabitants (Tier 2)</i> · <i>Stakeholder engagement</i> · <i>Design minimizes land acquisition and involuntary resettlement (Tier 2)</i>

* Project-specific historical data available

Samuel De Champlain Bridge

Project overview

The Samuel De Champlain Bridge corridor is a primary gateway to Montréal, located to the East of the city. It is a public-private partnership (PPP) between the Government of Canada and Signature on the St. Lawrence Group (SSL), a consortium of companies responsible for the design, construction, financing, operation, maintenance, and rehabilitation of the 3.4 km long new bridge across the St. Lawrence River that replaced the old Champlain Bridge. The Samuel De Champlain Bridge has a three-corridor design, including two three-lane corridors for vehicular traffic, a two-lane transit corridor designed to accommodate a light rail transit system now in service, and a multi-use path for pedestrians and cyclists. The project also encompasses a 470 m long Île-des-Soeurs Bridge, which includes three-lane corridors for vehicular traffic and a multi-use path, as well as the reconstruction and widening of two sections of the approach roads. The use of the PPP model was evaluated using a rigorous financial and technical analysis, an independent business case, and a value-for-money assessment, which demonstrated that a PPP would be the most cost-effective method to deliver the project within an accelerated timeline, resulting in estimated net present value savings of approximately 33.7% (\$1.747 billion dollars), compared to delivery of the project using the traditional method.

Project cost: CAD 4.474 billion (USD 3.372 billion)

Sponsor: Government of Canada (Infrastructure Canada and Public Services and Procurement Canada)

Private partner: Signature on the Saint-Lawrence Group

Status: Bridge operational since December 2018; Entire corridor completed in October 2019

QII Indicators selection

Identification of project objectives

The Samuel De Champlain Bridge corridor project is designed to achieve the following objectives:

1. Ensure continued safety and service in the long term.
2. Promote economic growth through improved connectivity.
3. Provide value for money for Canadians by creating long lasting infrastructure that meets high technical standards as well as the needs of users, achieves high quality at the lowest life cycle cost possible, and is consistent with the Government of Canada's vision of an appropriate risk transfer to the Private Partner.
4. Foster sustainable development and urban integration by considering the project's environmental and social context, implementing rigorous environmental monitoring and mitigation measures, building a bridge that complements Montréal's landscape, and promoting sustainable transportation by incorporating a dedicated public transit corridor and a safe and accessible multi-use path for pedestrians and cyclists.

Table 7 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting, transportation, and urban infrastructure annexes of the Compendium of QII Indicators.

Incorporating resource considerations

Data coverage for the project was assessed based on the publicly available information and project documents published on the Infrastructure Canada website.⁸ There is meaningful overlap between the data contained on the website (primarily related to project design) and the QII Indicators, as illustrated in Table 7. Further, an evaluation of the project, which is in its early stages, is expected to generate data for most of the indicators.

⁸ Infrastructure Canada, "Samuel de Champlain Bridge Corridor Project," <https://www.infrastructure.gc.ca/nbsl-npsl/index-eng.html>

Ensuring balanced representation of QII Principles

The project's objectives emphasize alignment with five of the six QII Principles. The objective of boosting economic growth through improved connectivity is aligned with Principle 1. Value for money, risk sharing with the private sector, and a commitment to integrity in the procurement process are captured in Principles 2 and 6. Sustainable development and urban integration objectives are aligned with Principles 1 and 3. Continued safety and service in the long term are aligned with Principle 4.

Table 7. QII indicator selection for the Samuel De Champlain Bridge

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Promote economic growth through improved connectivity	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	· <i>Average daily traffic (or infrastructure utilization rate)*</i> · <i>Total employment effects (Tier 2)</i> · Direct jobs created by the project (Operations and Maintenance) · Construction jobs (Temporary Construction)
Provide value for money	Principle 2: Raising economic efficiency in view of life-cycle cost	· <i>Risk mitigation measurement (Tier 2)*</i> · <i>Value-for-money analysis conducted (Tier 2)*</i> · Additional capacity per currency invested · Vehicle operating costs (roads/ports)
	Principle 6: Strengthening infrastructure governance	· <i>Transparency and accountability measures in procurement and financial management supported in implementation (Tier 2)*</i>
Foster sustainable development and urban integration	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	· <i>Access for non-motorized traffic (Tier 2)*</i>
	Principle 3: Integrating environmental considerations in infrastructure investments	· <i>GHG Emissions reduced/avoided</i> · <i>Adopts an environmental and social mitigation hierarchy framework*</i> · Local air pollutants reduced (Tier 2)* · Measures to limit noise pollution (Tier 2)* · Contribution to country's transport sector NDC target (Tier 2)
Ensure continued safety and service in the long term	Principle 4: Building resilience against natural disasters and other risks	· <i>Project introduces design or technical characteristics that enable operability in all weather conditions*</i> · <i>Diligent maintenance and continuity (Tier 2)*</i> · Design durable to disaster risks (Tier 2) · Budget committed to disaster and climate risk resilience measures (Tier 2)

* Indicator cited in the project documents

Mombasa Port

Project overview

The Mombasa Port Development project responded to the increasing demand for cargo handling and improved the efficiency of port operation at the Mombasa Port, one of the largest international trade ports in East Africa, by constructing a new container terminal and installing cargo handling equipment. Prior to the expansion, the port's capacity was limited to approx. 450,000 TEU per year, less than half of projected demand, and it needed to accommodate the increasing size of container ships, improve access to the port, and increase the efficiency of port operation. The expansion and modernization of the port, in conjunction with other transportation and logistics projects aimed at improving connectivity between Kenya and neighboring countries, has contributed to the promotion of trade and social and economic development not only in Kenya but across East Africa.

Project cost: JPY 31,735 million (USD 291 million)

Sponsor: Kenya Ports Authority

Financing: Japanese ODA loan to Kenya Ports Authority (guaranteed by the Government of Kenya) for 83% of the total project cost

Status: Operational since 2017

QII Indicators selection

Identification of project objectives

The primary objectives of the Mombasa Port Development project were to increase port capacity and improve performance of port infrastructure, ultimately contributing to economic and social development. The following environmental and social aspects objectives informed the design and implementation of the project: (i) minimize negative impacts on the natural environment; (ii) provide adequate compensation to the fishing communities affected by the port expansion; and (iii) prevent the spread of HIV/AIDS. Table 8 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting and transportation annexes of the Compendium of QII Indicators.

Incorporating resource considerations

The project was subject to Japanese ODA reporting requirements as well as an ex-post evaluation two years after completion. The evaluation, which was used to assess data coverage, included assessment of a wide range of outcomes with significant overlap with QII Indicators, as illustrated in Table 8.

Ensuring balanced representation of QII Principles

The objectives of the Mombasa Port project are closely aligned with Principles 1 and 2 since the expansion of the port's capacity and greater efficiency of operations were expected to deliver significant economic and social development benefits. The project also prioritized mitigation of environmental and social risks (Principles 3 and 5) during its construction and operations.

Table 8. QII indicator selection for the Mombasa Port project

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Expand port capacity and improve performance of port infrastructure	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	· <i>Passenger and freight volumes by mode of transport*</i> · <i>Berth moves (Tier 2)*</i> · <i>Container dwell time (Tier 2)*</i> · Direct jobs supported by the project (Operations and Maintenance)* · Direct jobs created by the project (Operations and Maintenance)* · Construction jobs (Temporary Construction) · Total employment effects (Tier 2) · Development of employee skills (Tier 2)
Improve the operational efficiency of the port	Principle 2: Raising economic efficiency in view of life-cycle cost	· <i>Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)*</i> · Additional capacity per currency invested* · Value-for-money analysis conducted (Tier 2) · Vehicle operating costs (roads/ports)
Minimize adverse impacts on the natural environment	Principle 3: Integrating environmental considerations in infrastructure investments	· <i>Adopts an environmental and social mitigation hierarchy framework*</i> · GHG Emissions reduced/avoided · Local air pollutants reduced (Tier 2) · Water consumption (conservation) · Energy consumption (conservation) · Provides tools, and data to improve E&S practices for stakeholder engagement (Tier 2) · Measures to limit noise pollution (Tier 2)
Provide adequate compensation to the fishing communities affected by the project and avoid the spread of HIV/AIDS	Principle 5: Integrating social considerations in infrastructure investment	· <i>Community development contributions</i> · <i>Compensation to displaced inhabitants (Tier 2)</i> · Stakeholder engagement · Design minimizes land acquisition and involuntary resettlement (Tier 2)

* Project-specific historical data available

Port of Taranto

Project overview

The project encompasses the modernization and the development of the back-port areas of the Port of Taranto, one of the core seaports of the Apulia region of Italy. The regional development strategy envisions the Port of Taranto providing a common logistics area and connection to a multimodal transportation network integrating the economies of two Italian regions, Apulia and Basilicata. The modernization of the Port of Taranto will take place alongside the creation of an Eco Industrial Park,⁹ a newly designed eco-sustainable production area integrated in the dry port area of the Port of Taranto and falling within the interregional Special Economic Zone (SEZ) of the Ionian Sea. The first phase of the Industrial Park implementation will include construction of primary infrastructure and access roads, followed by the second phase encompassing a railway, secondary roads, industrial equipment, warehouses, and green systems, among other works. The Eco Industrial parks will be “green” - both in terms of the performance of the infrastructure to be built and the orientation of the productive activities taking place there - and has been designed in line with the EU’s strategic measures for a “Greener Europe.” The construction of the Eco Industrial Park will facilitate the evolution of the Port of Taranto from a transshipment port, traditionally serving the steel industry, to a logistics hub integrated into supply chains for higher value-added goods.

Project cost: EUR 50 million for the first phase; TBD for the second phase

Sponsor: Ionian Sea Port System Authority

Private partner: to be selected

Financing: EUR 50 million from the Government of Italy through the National Recovery and Resilience Plan (Piano Nazionale di Ripresa e Resilienza – PNRR); a private operator will provide financing for the second phase through project financing or a Public Private Partnership

Status: Pre-operational

QII Indicators selection

Identification of project objectives

The primary objective of the Port of Taranto project is to enhance the economic attractiveness of the entire Taranto Ionian area on a national and international scale by strengthening the Port’s operations and accessibility, both physical and digital, reversing the downward trend in throughput by catering to companies in future growth sectors that are at the core of the regional development strategy - logistics, green economy, and blue economy. The Ionian Sea Port Authority intends to mobilize private financing and operational expertise for the construction and management the Port and the Eco Industrial Park. The project’s green design will contribute to sustainable development and ensure high levels of environmental performance and climate resilience. Table 9 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting and transportation annexes of the Compendium of QII Indicators.

Incorporating resource considerations

Information on availability of data for cross-cutting and transportation sector QII Indicators, as well as expected indicator values for the project, was provided by the Italian Ministry of Economy and Finance and shows meaningful overlap with QII Indicators, as illustrated in Table 9.

⁹ An eco-industrial park is an industrial area designed and managed sustainably to reduce the environmental impact of industrial activities. These parks promote synergy between businesses, encouraging the sharing of resources, energy efficiency and the adoption of green practices.

Ensuring balanced representation of QII Principles

The objectives of the Port of Taranto align with five of the six QII Principles. The Port's expected role in regional economic development and mobilization of private financing for the project are aligned with Principles 1; the focus on improvement of logistics performance is aligned with Principle 2; the focus on sustainable development includes environmental and climate resilience considerations under Principles 3 and 4; and the role of governance in attracting private investment is captured in Principle 6.

Table 9. QII indicator selection for the Port of Taranto project

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Increase Port utilization through integration with the Eco Industrial Park	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	• <i>Passenger and freight volumes by mode of transport*</i>
Mobilize private financing for the project		• <i>Private capital mobilization*</i> • <i>Business innovation (Tier 2)*</i> • Time between contract signature and financial close
	Principle 6: Strengthening infrastructure governance	• <i>Introduction of new type of concession terms (Tier 2)*</i> • Information disclosure (Tier 2) • Anti-corruption protocols & procedures (Tier 2) • Transparency and accountability measures in procurement and financial management supported in implementation (Tier 2) • Information transparency & access (Tier 2)*
Improve the Port's operational performance	Principle 2: Raising economic efficiency in view of life-cycle cost	• <i>Value-for-money analysis conducted (Tier 2)*</i> • <i>Risk mitigation measurement (Tier 2)*</i> • Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2) • Additional capacity per currency invested*
Ensure high level of environmental performance	Principle 3: Integrating environmental considerations in infrastructure investments	• <i>GHG Emissions reduced/avoided</i> • <i>Adopts an environmental and social mitigation hierarchy framework*</i> • <i>Sustainability certification*</i> • Local air pollutants reduced (Tier 2) • Energy consumption (conservation) • Water consumption (conservation) • Energy & water efficiency (Tier 2) • Measures to limit noise pollution (Tier 2)* • Provides tools, and data to improve E&S practices for stakeholder engagement (Tier 2)* • Contribution to country's transport sector NDC target (Tier 2)
Integrate climate change risks into the project design	Principle 4: Building resilience against natural disasters and other risks	• <i>Budget committed to disaster and climate risk resilience measures (Tier 2)*</i> • <i>Disaster risk financing (Tier 2)*</i> • Design durable to disaster risks (Tier 2)* • Improvement in employee technical and disaster management skills (Tier 2)

* Historical or expected value for the indicator included in the project documents

Solaria Castilian PV Plants

Project overview

The project involves the construction and operation of seven solar photovoltaic (PV) plants located in the Spanish regions of Castilla y Leon, Castilla-La Mancha and Extremadura. They will have a total capacity of 261 MW and generate around 455 GWh a year, which is equivalent to the power consumption of 121,300 Spanish homes. All the plants are located in less-developed regions of Spain (“cohesion regions”) prioritized under the EU Cohesion Policy, which seeks to correct imbalances between countries and regions and deliver on the Union's political priorities, especially the green and digital transition. The plants are owned by Solaria, which is also the developer, builder, and operator. The output of the plans will be sold through the wholesale market and unsubsidized commercial PPAs, without applying for any form of public subsidy or support scheme.

Project cost: EUR 157 million

Sponsor: Corona Borealis Fotovoltaica Slu

Financing: EUR 68 million in project finance from the European Investment Bank (EIB). The remaining amount of the project finance was underwritten by Natixis.

Status: Pre-operational

QII Indicators selection

Identification of project objectives

The construction of the PV parks will promote security of energy supply in Spain and support job creation, social cohesion, and economic recovery of areas that have been particularly affected by the COVID-19 pandemic. Additionally, it will contribute to Spain's 2030 targets for reducing CO2 emissions and to Spain's renewable energy targets set out in the National Energy and Climate Plan (PNEC), which calls for increasing the share of renewable energy sources in gross final energy consumption from 20% in 2020 to 42% in 2030. As a sizable renewable energy project developed without a public support scheme, the project is expected to make it easier for similar projects to raise financing in the future. Table 10 summarizes these objectives and maps them to QII Principles and applicable indicators from the cross-cutting and energy annexes of the Compendium of QII Indicators.

Incorporating resource considerations

Data coverage was assessed based on the information and documentation published on the EIB and InvestEU websites, which covers project design, procurement process, and expected impacts, and demonstrates significant overlap with QII Indicators, as illustrated in Table 10.¹⁰

Ensuring balanced representation of QII Principles

The project objectives align with three of the six QII Principles. The plants' contribution to job creation, social cohesion, and economic recovery is aligned with Principle 1; the need to achieve bankability without public support is aligned with Principle 2; and the contribution to energy supply security and reduction in CO2 emissions fall under Principle 3.

¹⁰ European Investment Bank, “Solaria Castilian PV Plants,” <https://www.eib.org/en/projects/all/20200100>. InvestEU, “Solaria Castilian PV Plants,” https://investeu.europa.eu/investeu-operations/solaria-castilian-pv-plants_en

Table 10. QII indicator selection for the Solaria Castilian PV Plants

Objective	QII Principle	Applicable indicators <i>Selected indicators</i>
Support job creation, social cohesion, and post-COVID-19 economic recovery	Principle 1: Maximizing the positive impact of infrastructure to achieve sustainable growth and development	· <i>Jobs created in renewable energy and other energy transition-related fields (Tier 2)*</i> · Direct jobs created by the project (Operations and Maintenance)* · Construction jobs (Temporary Construction)* · Development of employee skills (Tier 2) · Total employment effects (Tier 2)
Promote security of energy supply	Principle 3: Integrating environmental considerations in infrastructure investments	· <i>Renewable energy generated or installed*</i>
Reduce CO2 emissions and contribute to Spain's renewable energy targets		· <i>Contribution to country's energy sector NDC target (Tier 2)*</i> · <i>GHG emissions reduction/avoided*</i> · Adopts an environmental and social mitigation hierarchy framework*
Ensure project bankability without a public support scheme	Principle 2: Raising economic efficiency in view of life-cycle cost	· <i>Levelized cost of electricity (LCOE) (Tier 2)</i> · <i>Risk mitigation measurement (Tier 2)</i> · Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2) · Sustainable pricing mechanisms (Tier 2)

* Historical or expected value for the indicator included in the project documents

Emerging Lessons

Undertaking the pilot on the ten aforementioned projects has generated learnings on the practical aspects of using the Compendium and the Guidance Note, validated assumptions on data availability and the use of indicators by G20 members, and demonstrated specific ways in which the use of the QII Indicators could support quality infrastructure investment.

On the application of QII Indicators

The exercise appears to have validated that QII Indicators are broadly applicable across country and sector contexts and project development stages, as evidenced by the fact that the unique objectives of all pilot projects to date lend themselves to be successfully mapped to a set of QII Indicators. Several of the indicators were selected for more than one pilot project, demonstrating their cross-sectorial relevance. Indicator definitions proved to be sufficiently flexible to accommodate differences in country legal frameworks and sector-specific concerns. For example, although governance indicators on *Information disclosure* and/or *Transparency and accountability measures* apply to the Darwin Water Supply, São Simão, Samuel De Champlain Bridge, and Makassar-Parepare Railway projects, they would entail different actions for each of these projects, depending on applicable regulatory requirements. Considering this, countries may want to supplement the broad and flexible definitions provided in the Guidance Note for qualitative binary indicators such as those cited above with detailed, specific requirements based on national circumstances and best practices and standards referenced in the Guidance Note to ensure that the indicator can be measured accurately and consistently. Another way to avoid potential errors in measurement is to collect additional information on how the project meets indicator criteria in the form of a comment explaining the Y/N indicator value, as suggested in the Guidance Note.

The mapping of project objectives to QII Indicators validated that the Compendium can be used to reflect a wide range of expected project outcomes. In some cases, however, it required an additional step of identifying intermediate project outcomes that are expected to lead to the ultimate project objectives targeted by the stakeholders, because the Compendium relies on measurement of intermediate outcomes, rather than the final impact, for such objectives as local or regional growth and development (e.g. expected from the Delhi MRTS, the Mombasa Port and other projects).

The exercise also suggests that the categorization of indicators in the Compendium under specific QII Principles and sectors should be viewed as a guideline rather than a strict rule. The scope of the pilot projects did not always fall under one sector: for instance, while most indicators for the Nhat Tan Bridge and Connecting Road and the Samuel De Champlain Bridge can be found in the Transportation annex of the Compendium, which covers roads, these are also urban projects. The importance of the municipality's capacity to operate and maintain the Nhat Tan Bridge and Connecting Road led to the selection of *Diligent maintenance and continuity* from the Urban Infrastructure annex. Similarly, *Access for non-motorized traffic*, an Urban Infrastructure indicator, is suggested for the Samuel De Champlain Bridge given its objective to improve sustainability of transport and urban integration in Montréal. The São Simão concession covering three distinct sectors also necessitated the use of indicators from across the Compendium with some indicators applying to the concession as a whole and others to one of its components. Some judgment may also occasionally be needed in mapping specific indicators to QII Principles: while the *Development of Employee Skills* indicator is categorized under Principle 1 in the Compendium, in cases like the Delhi metro and the Nhat Tan Bridge and Connecting Road, employee training specifically centered on proper maintenance and operations, which falls under Principle 4.

The pilot projects revealed uneven data coverage across indicators, which was anticipated and accounted for through tiering of indicators in the Compendium. The limited number of operational pilot projects (six of the ten analyzed to date) does not allow for definitive conclusions regarding indicator tiering; however, some initial lessons on coverage across QII Principles can be drawn. Much of the quantitative data required for operational and financial indicators (related to Principles 1 and 2) was available in project documents for operational projects. Values for many of the qualitative binary indicators under Principles 1, 2, and especially for Principle 6 covering governance, could also be determined from the project information, even if such indicators were not explicitly tracked for the projects. Environmental output or outcome indicators (Principle 3) were generally tracked (or are expected to be tracked if they are in pre-operational stages) for projects associated with significant environmental impacts. Pilot projects demonstrate that social considerations (Principle 5) are commonly incorporated into project development plans and into countries' legal frameworks through stakeholder consultations and provisions to compensate displaced inhabitants. None of the projects appear to collect disaggregated data to analyze impacts on underserved groups, which may have been salient for some, particularly the Delhi MRTS project. Data on and tracking of indicators related to climate resilience also proved rare in the available documentation. It is important to note that the Compendium primarily offers output (or design) indicators to track alignment with Principle 4, precisely because few commonly used outcome (or results) indicators are available to measure resilience. As countries prepare to face enormous climate adaptation challenges in the coming years, it will be important to set out credible ways of monitoring and measuring adaptation and resilient infrastructure solutions, following the principle "what gets measured gets done."

On the potential uses of QII Indicators

The findings of the exercise point to several promising opportunities to leverage QII Indicators to maximize the positive impacts of infrastructure investment.

QII Indicators can help draw stakeholders' attention to performance in areas that tend to be subject to less measurement and monitoring and provide a means to systematically track them. While indicators related to operational and financial data (mostly under Principles 1 and 2) are almost always tracked given their importance for project financial returns, indicators reflecting environmental, social and governance performance are less commonly required and tracked by project stakeholders. A balanced set of QII Indicators can help shift this practice by giving prominence to all applicable QII Principles, increasing the chances that non-financial aspects of project quality will be prioritized alongside the financial ones.

While most pilot projects nominated by G20 members for this exercise are large and ambitious in their scope and aim to fulfill multiple objectives, it is important to note that this does not need to be the case for an infrastructure project to be aligned with QII Principles. In fact, the proliferation of project objectives can in some cases be counterproductive to achieving impact because it disperses the attention of stakeholders, in addition to making monitoring more complex and costly. A smaller, more narrowly focused project may also demonstrate alignment with QII principles by tracking indicators that assess performance with respect to its one primary objective, supplemented by indicators related to its management of material risks and quality considerations that need to be considered in project design and implementation.

QII Indicators can facilitate system-level planning when they form part of a common approach to incorporating quality considerations into project development. The Delhi MRTS and the Nhat Tan Bridge and Connecting Road projects demonstrate that applying comparable indicators across parts of the transportation network can help decision-makers analyze linkages between modes of transport and between interconnected transport infrastructure projects. Similarly, the designs of the Darwin Water Supply and the São Simão projects rely on data related to different parts of the water supply system, including intake, treatment, storage and distribution. The adoption of QII

Indicators by government planning agencies could contribute to ensuring the consistency and completeness of this data.

The measurement of QII Indicators over time can also ensure that alignment with QII Principles is maintained across project phases, including if the project is reconfigured or expanded as was the case in several of the pilot projects. For instance, the Mombasa Port underwent changes due to the suspension of plans to privatize it, increases in the cost of equipment, and a new railway connection to the port becoming available. The Delhi MRTS project design was modified to connect Line 2 to the Commonwealth Games venues. In these cases, the application of QII Indicators could help stakeholders evaluate how the changes may affect different aspects of project quality.

Collecting and disclosing data on project results more broadly can help set realistic expectations regarding the results that can be achieved through a particular type of investment. Notwithstanding data gaps with respect to some QII Principles, the pilot projects demonstrate widespread use of baselines and targets to benchmark expected impacts and appraise investment decisions. In the long run, supplementing monitoring with ex-post evaluation to verify the achievement of expected impacts and draw lessons, as was done in the case of Japanese ODI-financed projects (Delhi MRTS, Nhat Tan Bridge and Connecting Road, and Mombasa Port), enables learning and continuous innovation and improvement in infrastructure project design.

Ultimately, the adoption of QII Indicators can pave the way for the diversification of instruments and sources of financing for infrastructure by improving data availability on a broad range of project outcomes and giving comfort to financiers regarding project quality. Firstly, the harmonization of measurement frameworks using the common definitions provided by the QII Indicators can help generate credible and consistent data reporting across infrastructure projects, facilitating appraisal of investments. Secondly, the adoption of QII Indicators for project monitoring can serve as a first step in creating feedback loops that help improve the quality of subsequent projects. Lastly, reliable data on project performance across environmental, social and governance dimensions captured by the QII Indicators can improve companies' access to financing from impact-focused financiers, and allow them to utilize results-based financing, including sustainability-linked loans and bonds.

ANNEX: Proposed QII Indicators

São Simão Water, Sanitation, and Solid Waste Concession

QII Principle / objective	Principle 1 / Universalization of access to water, wastewater, and solid waste management services
Indicator	Population with access to safe drinking water
Unit	%
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	87.7% in 2021 ¹¹
Indicator	Population with access to improved sanitation
Unit	%
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	81.30% in 2021 ¹²
Indicator	Proportion of municipal solid waste (MSW) collected and managed in controlled facilities
Unit	%
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 1 / Job creation and improvement in incomes and working conditions
Indicator	Total employment effects (Tier 2)
Unit	Number of full-time equivalent jobs
Source	Calculated based on project data
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 2 / Reduction in per capita water use and losses
Indicator	Sustainable pricing mechanisms
Unit	Y/N
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Non-revenue water (commercial)
Unit	%
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	N.A. (Commercial and technical totaled 44.0% in 2021) ¹³
Indicator	Non-revenue water (technical)
Unit	%
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	N.A. (Commercial and technical totaled 44.0% in 2021)
QII Principle / objective	Principle 2 / Financial sustainability of sanitation services at affordable end-user tariffs
Indicator	Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)
Unit	%
Source	Project financial model
Timing and frequency	Once during the design stage; annually or as needed during the operational stage
Historical values	N.A.
QII Principle / objective	Principle 3 / Adequate wastewater treatment capacity and quality
Indicator	Wastewater treated
Unit	m ³
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 3 / Increase in share of waste recycled
Indicator	Total weight of waste diverted from disposal and its recovery
Unit	tonnes
Source	Project operations reports; sector regulator; project KPI reporting
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 4 / Reliability of water and wastewater services
Indicator	Preventive maintenance (Tier 2)
Unit	%

¹¹ “Estudo de Engenharia, Logística e Afins: Água e Esgoto,” 2021, p. 34.

¹² Ibid, p. 60.

¹³ Ibid, p. 38.

Source	Project financial documents
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Ruptures in transportation and distribution pipes (Tier 2)
Unit	Number / km
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 6 / Implementation of the concession in accordance with applicable laws and regulations
Indicator	Information disclosure (Tier 2)
Unit	Y/N
Source	Project documents
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y in 2021 ¹⁴
Indicator	Transparency and accountability measures in procurement and financial management supported in implementation (Tier 2)
Unit	Y/N
Source	Assessment based on financial and operational reports, project legal agreements
Timing and frequency	Annually during pre-operational and operational stages
Historical values	N.A.

¹⁴ São Simão Municipality, “Concorrência Pública 003/2021,” <https://saosimao.go.gov.br/tp/concorrenca-publica-003-2021/>.

Darwin Region Water Supply

QII Principle / objective	Principle 2 / Select the most cost effective and efficient long-term water supply option
Indicator	Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)
Unit	%
Source	Project financial model
Timing and frequency	Once during the design stage; annually or as needed during the operational stage
Historical values	N.A.
Indicator	Value-for-money analysis conducted (Tier 2)
Unit	Y/N
Source	Project appraisal documents
Timing and frequency	Once during pre-operational phase
Historical values	Y ¹⁵
QII Principle / objective	Principle 3 / Minimize adverse environmental and social impacts
Indicator	GHG Emissions reduced/avoided
Unit	Tonnes per year
Source	Project environmental impact assessment
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Adopts an environmental and social mitigation hierarchy framework
Unit	Y/N
Source	Project social and environmental assessment
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y ¹⁶
Indicator	Provides tools, and data to improve E&S practices for stakeholder engagement (Tier 2)
Unit	Y/N
Source	Project appraisal documents, stakeholder engagement plan
Timing and frequency	Once during pre-operational stage
Historical values	Y ¹⁷
QII Principle / objective	Principle 4 / Provide safe, reliable, and sustainable water supply, considering climate change and water demand changes
Indicator	Design durable to disaster risks (Tier 2)
Unit	Y/N
Source	Project design documents, insurance documents
Timing and frequency	Once during pre-operational stage
Historical values	N.A.
Indicator	Design improves resilience to failure of critical links (Tier 2)
Unit	Y/N
Source	Project design documents
Timing and frequency	Once during pre-operational stage
Historical values	Y ¹⁸
QII Principle / objective	Principle 5 / Consider interests of diverse stakeholders in project design
Indicator	Stakeholder engagement
Unit	Y/N
Source	Project appraisal and public consultation documents
Timing and frequency	Once during pre-operational stage
Historical values	Y ¹⁹
QII Principle / objective	Principle 6 / Thoroughly assess and monitor implementation of a nationally significant, large-scale project
Indicator	Information disclosure (Tier 2)
Unit	Y/N
Source	Project documents
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y ²⁰
Indicator	Transparency and accountability measures in procurement and financial management supported in implementation (T. 2)
Unit	Y/N
Source	Assessment based on financial and operational reports, project legal agreements
Timing and frequency	Annually during pre-operational and operational stages
Historical values	N.A.

¹⁵ PWC, "Adelaide River Off-stream Water Storage Project: Detailed Business Case Key Findings."

¹⁶ Department of Trade, Business and Innovation, "Adelaide River Off-stream Water Storage - Part A: Preliminary Assessment," 2020.

¹⁷ Ibid.

¹⁸ PWC, "Adelaide River Off-stream Water Storage Project: Detailed Business Case Key Findings."

¹⁹ Department of Trade, Business and Innovation, "Adelaide River Off-stream Water Storage - Part A: Preliminary Assessment," 2020.

²⁰ Ibid.

Sarnese Vesuvian District Water Supply²¹

QII Principle / objective	Principle 1 / Maximize socio-economic benefits in the implementation of the project
Indicator	Total employment effects (Tier 2)
Unit	Number of full-time equivalent jobs
Source	Calculated based on project data
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 1 / Improve system performance
Indicator	Piped water pressure (Tier 2)
Unit	# seconds/liter
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 2 / Reduce system losses
Indicator	Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)
Unit	%
Source	Project financial model
Timing and frequency	Once during the design stage; annually or as needed during the operational stage
Historical values	N.A.
Indicator	Non-revenue water (technical)
Unit	%
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 3 / Decrease energy consumption in the water network and associated CO2 emissions
Indicator	Energy consumption (conservation) (Tier 2)
Unit	kWh
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Level of water stress (Tier 2)
Unit	%
Source	Environmental or water sector regulatory agencies
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 3 / Reuse, recycle, or recover at least 70% of construction and demolition waste
Indicator	Total weight of waste diverted from disposal and its recovery
Unit	Tonnes
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 4 / Contribute to sustainable use of water resources and resilience of supply
Indicator	Ruptures in transportation and distribution pipes (Tier 2)
Unit	Number / km
Source	Project operations reports, sector regulator
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 5 / Contribute to sustainable use of water resources and resilience of supply
Indicator	Multi-generational equity (Tier 2)
Unit	Y/N
Source	Project design documents
Timing and frequency	Once during pre-operational stage
Historical values	Y
QII Principle / objective	Principle 6 / Continue to advance digitalization of the water supply system
Indicator	Updated, timely shared, consistent, and comparable water information systems (Tier 2)
Unit	Y/N
Source	Project design documents; project information system
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y

²¹ Sarnese Vesuvian District Water Supply project description and indicator data provided by the Italian Ministry of Economy and Finance for the pilot exercise.

Makassar-Parepare Railway²²

QII Principle / objective	Principle 1 / Contribute to economic growth and job creation in South Sulawesi
Indicator	Total employment effects (Tier 2)
Unit	Number of full-time equivalent jobs
Source	Calculated based on project data
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 1 / Facilitate transport of goods and passengers
Indicator	Passenger and freight volumes by mode of transport
Unit	Passenger-km, tonne-km
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principles 1 and 2 / Maximize efficiency of use of public funds and transfer portion of the risk to the private sector
Indicator	Private capital mobilization
Unit	Currency (IDR)
Source	Project financial documents
Timing and frequency	Depending on financing and/or refinancing timing and frequency
Historical values	IDR 693.83 billion syndicated loan financing arranged by BSI in 2021
Indicator	Additional capacity per currency invested
Unit	Passenger-kms / currency; tons / currency
Source	Project financial model
Timing and frequency	Once during pre-operational phase
Historical values	N.A.
Indicator	Risk mitigation measurement (Tier 2)
Unit	Y/N
Source	Project due diligence documents; project financial model
Timing and frequency	Once during pre-operational stage
Historical values	N.A.
Indicator	Value-for-money analysis conducted (Tier 2)
Unit	Y/N
Source	Project appraisal documents
Timing and frequency	Once during pre-operational phase
Historical values	Y
QII Principle / objective	Principle 3 / Provide a more sustainable alternative to road transportation
Indicator	GHG emissions reduced / avoided
Unit	Tonnes per year
Source	Estimated based on project utilization, traffic studies
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 5 / Maximize benefits and minimizes adverse impacts on affected communities
Indicator	Stakeholder engagement
Unit	Y/N
Source	Project appraisal and public consultation documents
Timing and frequency	Once during pre-operational stage
Historical values	N.A.
Indicator	Design minimizes land acquisition and involuntary resettlement (Tier 2)
Unit	Y/N
Source	Project design documents; social and environmental assessment
Timing and frequency	Once during pre-operational stage
Historical values	N.A.
QII Principle / objective	Principle 6 / Ensure transparency and proper use of funds
Indicator	Transparency and accountability measures in procurement and financial management supported in implementation (Tier 2)
Unit	Y/N
Source	Assessment based on financial and operational reports, project legal agreements
Timing and frequency	Annually during pre-operational and operational stages
Historical values	N.A.
Indicator	Anti-corruption protocols & procedures (Tier 2)

²² Makassar-Parepare Railway presentation provided by Indonesia for the pilot exercise.

Unit	Y/N
Source	Project documents
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y
Indicator	Information disclosure (Tier 2)
Unit	Y/N
Source	Project documents
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y

Delhi MRTS²³

QII Principle / objective	Principle 1 / Reduce traffic congestion and travel time
Indicator	Average daily ridership
Unit	People per day
Source	Project operations reports
Timing and frequency	Daily during the operational stage
Historical values	Target: 2,980,000 in 2013. Actual: 1,930,000 in 2013, 2,360,000 in 2015
Indicator	Decreased journey times
Unit	Minutes per trip
Source	Central Road Research Institute
Timing and frequency	Annually during operational stage
Historical values	30 minutes per trip
Indicator	Journey experience
Unit	Various
Source	Passenger survey
Timing and frequency	Annually during operational stage
Historical values	Survey results on various topics available in ex-post evaluation
QII Principle / objective	Principle 2 / Ensure efficient operation of the transport system
Indicator	Internal rate of return (IRR) and/or Economic rate of return (ERR)
Unit	%
Source	Project financial model
Timing and frequency	Once during the design stage; annually or as needed during the operational stage
Historical values	FRR 3.84%; ERR 13.72% in 2010, including the final scope; ERR 25.61% at the time of the ex-post evaluation
QII Principle / objective	Principle 3 / Reduce energy consumption, air pollution, and noise pollution
Indicator	GHG Emissions reduced/avoided
Unit	Tonnes per year
Source	Estimated based on project utilization, traffic studies
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Local air pollutants reduced (Tier 2)
Unit	Tonnes per year
Source	Project environmental impact assessment; estimates from city-level measurements
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Measures to limit noise pollution (Tier 2)
Unit	Y/N
Source	Project design documents
Timing and frequency	Once during the pre-operational stage
Historical values	Y
QII Principle / objective	Principles 1 and 4 / Provide reliable service
Indicator	Development of employee skills (Tier 2)
Unit	Number of employees receiving training
Source	SMRC training institute
Timing and frequency	Annually during pre-operational and operational stages
Historical values	About 8,000 people taking courses at the institute annually
Indicator	Diligent maintenance and continuity (Tier 2)
Unit	Y/N
Source	Project design documents, operational manuals and reports
Timing and frequency	Once during the pre-operational stage
Historical values	Y
QII Principle / objective	Principle 5 / Ensure that public transit is convenient for all passengers
Indicator	Usability of public transport, by gender (Tier 2)
Unit	%
Source	Passenger survey
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Proportion of population that has convenient access to public transport, by sex, age, and disabilities (Tier 2)
Unit	%
Source	City-level assessment of access to public transport
Timing and frequency	Annually during operational stage
Historical values	N.A.

²³ “Delhi Mass Rapid Transport System Project Phase 2 (I) - (V),” 2015 Ex-Post Evaluation of Japanese ODA Loan Project.

Nhat Tan Bridge and Connecting Road²⁴

QII Principle / objective	Principle 1 / Alleviate congestion and improve efficiency of logistics
Indicator	Average daily traffic (or infrastructure utilization rate)
Unit	# vehicles per year
Source	Project operations documents, traffic survey
Timing and frequency	Annually during operational stages
Historical values	Bridge: 69,257 passenger car units (PCU)/day; Road: 15,153 PCU/day in 2020
Indicator	Journey times
Unit	Minutes
Source	Traffic survey
Timing and frequency	Annually during operational stages
Historical values	16-22 minutes from Nhat Tan Bridge to Noi Bai International Airport via the Connecting Road
QII Principle / objective	Principle 1 / Ensure long-term benefits of the project through proper operation and maintenance
Indicator	Development of employee skills (Tier 2)
Unit	Number
Source	Report from the institution providing employee training
Timing and frequency	Annually during pre-operational and operational stages
Historical values	N.A.
Indicator	Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)
Unit	%
Source	Project financial model
Timing and frequency	Once during the design stage; annually or as needed during the operational stage
Historical values	Nhat Tan Bridge ERR: appraisal 17.91% (Phase I), 21.36% (Phase III), ex-post 23.49% Connecting Road ERR: appraisal 17.25% (Phase I and II), ex-post 22.74%
Indicator	Vehicle operating costs (roads/ports)
Unit	Currency (VND) per km
Source	Project operations documents, traffic survey
Timing and frequency	Annually during operational stages
Historical values	Cost savings of 3,492.50 billion VND/year in 2017 for the bridge; 2,595.2 VND/year in 2020 for the connecting road
QII Principle / objective	Principle 4 / Ensure long-term benefits of the project through proper operation and maintenance
Indicator	Diligent maintenance and continuity (Tier 2)
Unit	Y/N
Source	Project design documents, operational manuals and reports
Timing and frequency	Once during the pre-operational stage
Historical values	Y
QII Principle / objective	Principle 3 / Mitigate environmental and social risks associated with bridge and road construction
Indicator	GHG Emissions reduced/avoided
Unit	Tonnes per year
Source	Estimated based on traffic studies
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Adopts an environmental and social mitigation hierarchy framework
Unit	Y/N
Source	Project social and environmental assessment
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y
QII Principle / objective	Principle 4 / Provide an alternate route to the airport in the event of a disaster
Indicator	Project introduces design or technical characteristics that enable operability in all weather conditions
Unit	Y/N
Source	Project design documents
Timing and frequency	Once during pre-operational stage
Historical values	Y
QII Principle / objective	Principle 5 / Comply with relevant laws and regulations on resettlements and land acquisition
Indicator	Compensation to displaced inhabitants (Tier 2)
Unit	Currency
Source	Project social and environmental assessment, financial reporting
Timing and frequency	Once. Timing depends on when compensation is expected to be provided.
Historical values	N.A.

²⁴ “Nhat Tan Bridge (Vietnam-Japan Friendship Bridge) Construction Project (I) (II) (III) and Noi Bai International Airport to Nhat Tan Bridge Connecting Road Construction Project (I) (II),” 2019 Ex-Post Evaluation of Japanese ODA Loan Project.

Samuel De Champlain Bridge²⁵

QII Principle / objective	Principle 1 / Promote economic growth through improved connectivity
Indicator	Average daily traffic (or infrastructure utilization rate)
Unit	# vehicles per year
Source	Project operations documents, traffic survey
Timing and frequency	Annually during operational stages
Historical values	50 million vehicles per year
Indicator	Total employment effects (Tier 2)
Unit	Number of full-time equivalent jobs
Source	Calculated based on project data
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 2 / Provide value for money
Indicator	Risk mitigation measurement (Tier 2)
Unit	Y/N
Source	Project due diligence documents; project financial model
Timing and frequency	Once during pre-operational stage
Historical values	Y
Indicator	Value-for-money analysis conducted (Tier 2)
Unit	Y/N
Source	Project appraisal documents
Timing and frequency	Once during pre-operational phase
Historical values	Y
QII Principle / objective	Principle 6 / Provide value for money
Indicator	Transparency and accountability measures in procurement and financial management supported in implementation (Tier 2)
Unit	Y/N
Source	Assessment based on financial and operational reports, project legal agreements
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y
QII Principle / objective	Principle 1 / Foster sustainable development and urban integration
Indicator	Access for non-motorized traffic (Tier 2)
Unit	Y/N
Source	Project design documents
Timing and frequency	Once during pre-operational phase
Historical values	Y
QII Principle / objective	Principle 3 / Foster sustainable development and urban integration
Indicator	GHG Emissions reduced/avoided
Unit	Tonnes per year
Source	Estimated based on traffic studies
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Adopts an environmental and social mitigation hierarchy framework
Unit	Y/N
Source	Project social and environmental assessment
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y
QII Principle / objective	Principle 4 / Ensure continued safety and service in the long term
Indicator	Project introduces design or technical characteristics that enable operability in all weather conditions
Unit	Y/N
Source	Project design documents
Timing and frequency	Once during pre-operational stage
Historical values	Y
Indicator	Diligent maintenance and continuity (Tier 2)
Unit	Y/N
Source	Project design documents, operational manuals and reports
Timing and frequency	Once during the pre-operational stage
Historical values	Y

²⁵ Infrastructure Canada, "Samuel de Champlain Bridge Corridor Project," <https://www.infrastructure.gc.ca/nbsl-npsl/index-eng.html>

Mombasa Port²⁶

QII Principle / objective	Principle 1 / Expand port capacity and improve performance of port infrastructure
Indicator	Passenger and freight volumes by mode of transport
Unit	TEU
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	1,416,000 TEU in 2019
Indicator	Berth moves (Tier 2)
Unit	Containers / hour
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	7 containers / hr (Berth 20); 46 containers / hr (Berth 21) in 2019
Indicator	Container dwell time (Tier 2)
Unit	Hours
Source	Northern Corridor Transport Observatory
Timing and frequency	Annually during operational stage
Historical values	95 hrs in 2019
QII Principle / objective	Principle 2 / Improve the operational efficiency of the port
Indicator	Internal rate of return (IRR) and/or Economic rate of return (ERR) (Tier 2)
Unit	%
Source	Project financial model
Timing and frequency	Once during the design stage; annually or as needed during the operational stage
Historical values	ERR: appraisal 12.1%, ex-post eval 8.2%; IRR: appraisal 7.5%, ex-post eval 6.6%
QII Principle / objective	Principle 3 / Minimize adverse impacts on the natural environment
Indicator	Adopts an environmental and social mitigation hierarchy framework
Unit	Y/N
Source	Project social and environmental assessment
Timing and frequency	Annually during pre-operational and operational stages
Historical values	Y
QII Principle / objective	Principle 5 / Provide adequate compensation to the affected fishing communities and avoid spread of HIV/AIDS
Indicator	Community development contributions
Unit	Currency
Source	Project social and environmental assessment, financial reporting
Timing and frequency	Once or annually. Timing depends on the nature of contributions and when they are expected to be provided.
Historical values	N.A.
Indicator	Compensation to displaced inhabitants (Tier 2)
Unit	Currency
Source	Project social and environmental assessment, financial reporting
Timing and frequency	Once. Timing depends on when compensation is expected to be provided.
Historical values	N.A.

²⁶ "Mombasa Port Development Project," 2019 Ex-Post Evaluation of Japanese ODA Loan Project.

Port of Taranto²⁷

QII Principle / objective	Principle 1 / Increase Port utilization through integration with the Eco Industrial Park
Indicator	Passenger and freight volumes by mode of transport
Unit	Number of passengers, TEU
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 1 / Mobilize private financing for the project
Indicator	Private capital mobilization
Unit	Currency
Source	Project financial documents
Timing and frequency	Depending on financing and/or refinancing timing and frequency
Historical values	N.A.
Indicator	Business innovation (Tier 2)
Unit	Y/N
Source	Project financial documents
Timing and frequency	Depending on financing and/or refinancing timing and frequency
Historical values	N.A.
QII Principle / objective	Principle 6 / Mobilize private financing for the project
Indicator	Introduction of new type of concession terms (Tier 2)
Unit	Y/N
Source	Project social and environmental assessment
Timing and frequency	Annually during pre-operational and operational stages
Historical values	N.A.
QII Principle / objective	Principle 2 / Improve the Port's operational performance
Indicator	Value-for-money analysis conducted (Tier 2)
Unit	Y/N
Source	Project appraisal documents
Timing and frequency	Once during pre-operational phase
Historical values	Y
Indicator	Risk mitigation measurement (Tier 2)
Unit	Y/N
Source	Project due diligence documents; project financial model
Timing and frequency	Once during pre-operational stage
Historical values	N.A.
QII Principle / objective	Principle 3 / Ensure high level of environmental performance
Indicator	GHG Emissions reduced/avoided
Unit	Tonnes per year
Source	Project environmental impact assessment
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Adopts an environmental and social mitigation hierarchy framework
Unit	Y/N
Source	Project social and environmental assessment
Timing and frequency	Annually during pre-operational and operational stages
Historical values	N.A.
Indicator	Sustainability certification
Unit	Y/N
Source	Project social and environmental assessment
Timing and frequency	Annually during pre-operational and operational stages
Historical values	N.A.
QII Principle / objective	Principle 4 / Integrate climate change risks into the project design
Indicator	Budget committed to disaster and climate risk resilience measures (Tier 2)
Unit	Currency
Source	Project financial documents
Timing and frequency	Depending on financing and/or refinancing timing and frequency
Historical values	N.A.
Indicator	Disaster risk financing (Tier 2)
Unit	Currency
Source	Project financial documents
Timing and frequency	Depending on financing and/or refinancing timing and frequency
Historical values	N.A.

²⁷ Port of Taranto project description and indicator data provided by the Italian Ministry of Economy and Finance for the pilot exercise.

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QII Principle / objective	Principle 1 / Promote energy supply security, job creation, social cohesion, and post-COVID-19 economic recovery
Indicator	Jobs created in renewable energy and other energy transition-related fields (Tier 2)
Unit	Number of full-time equivalent jobs
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	N.A.
QII Principle / objective	Principle 2 / Ensure project bankability without a public support scheme
Indicator	Levelized cost of electricity (LCOE) (Tier 2)
Unit	Currency per GWh
Source	Project financial model
Timing and frequency	Once during the design stage; annually or as needed during the operational stage
Historical values	N.A.
Indicator	Risk mitigation measurement (Tier 2)
Unit	Y/N
Source	Project due diligence documents; project financial model
Timing and frequency	Once during pre-operational stage
Historical values	N.A.
QII Principle / objective	Principle 3 / Reduce CO2 emissions and contribute to Spain's renewable energy targets
Indicator	Renewable energy generated or installed
Unit	GWh or MW
Source	Project operations reports
Timing and frequency	Annually during operational stage
Historical values	N.A.
Indicator	Contribution to country's energy sector NDC target (Tier 2)
Unit	Y/N
Source	Project design documentation
Timing and frequency	Once during pre-operational stage
Historical values	γ ²⁹
Indicator	GHG Emissions reduced/avoided
Unit	Tonnes per year
Source	Project environmental impact assessment
Timing and frequency	Annually during operational stage
Historical values	N.A.

²⁸ European Investment Bank, "Solaria Castilian PV Plants," <https://www.eib.org/en/projects/all/20200100>. InvestEU, "Solaria Castilian PV Plants," https://investeu.europa.eu/investeu-operations/solaria-castilian-pv-plants_en

²⁹ InvestEU, "Scoreboard – Solaria Castilian PV Plants," https://investeu.europa.eu/investeu-operations/solaria-castilian-pv-plants_en